Precalculus Review for Calculus

Functions and composition

1 Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:

a $(f \circ g)(x)$

b $(g \circ f)(x)$

c the domain of $f \circ g$

2 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.

Lines, exponentials, and trig — the toolkit calculus builds on

3 Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.

4 Solve $2e^{3x} = 14$ exactly and to three decimal places.

5 State the exact value of each expression, needed frequently when differentiating trig functions:

a $\sin\frac{\pi}{4}$ **b** $\cos\frac{\pi}{3}$ **c** $\tan\frac{\pi}{6}$